Aytaç Karabay, Ph.D.
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EXPERIENCE

Assistant professor, University of Birmingham Dubai, Dubai, UAE 2025–Present

Visiting scholar, New York University, Abu Dhabi, UAE 2025–2026

Adjunct assistant professor (part-time), Zayed University, Abu Dhabi, UAE 2024–2025

Postdoctoral researcher, New York University, Abu Dhabi, UAE 2022–2025
Supervisor: Dr. Daryl Fougnie

Postdoctoral researcher, University of Groningen, Groningen, NL 2019–2022
Supervisor: Prof. Elkan G. Akyürek

Research assistant, CUNY-Brooklyn College, New York, USA 2014–2015
Supervisor: Dr. Daniel Kurylo

Courses Taught

Research Methods C	Module lead	University of Birmingham	2025; 2026
The Neurobiology of Mental Illness	Lecturer	University of Birmingham	2025; 2026
Research Methods B	Module lead	University of Birmingham	2025; 2026
Research Methods D	Module lead	University of Birmingham	2025; 2026
Ideas, Concepts, and Ethics in Psychology and Neuroscience	Guest lecturer	University of Birmingham	2025; 2026
Introduction to Psychology	Module lead	Zayed University	2024
Introduction to Psychology	Teaching assistant	New York University|AD	2023
Thinking	Teaching assistant	New York University|AD	2023
Cognition and Attention	Guest lecturer	University of Groningen	2021; 2022
Academic Skills (2 cohorts)	Module lead	University of Groningen	2018–2019

Mentorship and supervision

Bachelors Thesis (x18)	Supervisor	University of Birmingham	2024–2026
Capstone Project (x2)	Mentor	New York University|AD	2022–2025
PhD Thesis (x1)	Supervisor	University of Groningen	2020–2026
Master's Thesis (x3)	Supervisor	University of Groningen	2017–2022
Bachelor's Thesis (x3)	Mentor	University of Groningen	2017–2022
Honors Research Practicum (x3)	Mentor	University of Groningen	2017–2018

EDUCATION

Ph.D., Cognitive Neuroscience, University of Groningen, Groningen, NL 2020
Supervisor: Prof. Elkan G. Akyürek

M.A., Experimental Psychology, CUNY-Brooklyn College, New York, USA 2015
Supervisor: Dr. Daniel D. Kurylo

B.A., Psychology, Ankara University, Ankara, TR 2011

PUBLICATIONS

Student trainees are underlined
** indicates shared first authorship*

Preprints and submitted papers

Karabay, A., Vriens, P., Adams, J., Tran, N., & Sahan, M. I. How you encode is how you remember: Center-to-surround inhibition during encoding underlies repulsion bias in working memory. *[under review]* 2025

Chen, I., Mathôt, S., van der Mijn, R., **Karabay, A.,** Bowman, H., & Akyürek, E. G. Concealed personally familiar face detection with EEG in rapid serial visual presentation. https://doi.org/10.31234/osf.io/axp4w_v1 2025

Chen, I. *, Buchel, P. *, **Karabay, A**., van der Mijn, R., Mathôt, S., & Akyürek, E. G. Concealed information detection in rapid serial visual presentation with oculomotor measures. https://doi.org/10.31234/osf.io/rtyu6 2023

Wang, S., **Karabay, A.**, & Akyürek, E. G. Evaluating Working Memory Representations during the Attentional Blink: A Comparative Modeling Approach. https://doi.org/10.1101/2022.09.15.508098 2022

Fougnie, D. L., **Karabay, A.**, Alvarez, G., & Brady, T. F. If at first you don't succeed, try, try again: Second chances reveal more information in working memory. https://doi.org/10.31234/osf.io/evqtn 2022

Journal papers

Wang, S., van der Mijn, R., **Karabay, A.**, & Akyürek, E. G. Multimodal detection of concealed information in rapid serial visual presentation using EEG and pupillometry. *Neuropsychologia , 225*, 109418. https://doi.org/10.1016/j.neuropsychologia.2026.109418 2026

Karabay, A., Wolff, M. J., Ruuskanen, V., & Akyürek, E. Behaviorally irrelevant feature matching increases neural and behavioral working memory readout. *Psychophysiology, 62(2).* https://doi.org/10.1111/psyp.70020 2025

Akyürek, E. G., Altınok, A., & **Karabay, A.** Concurrent consumption of cocoa flavanols and caffeine does not acutely modulate working memory and attention. *European Journal of Nutrition, 64*(35). 1–17. https://doi.org/10.1007/s00394-024-03514-8 2025

Karabay, A.*, Nijenkamp, R.*, Sarampalis, A., & Fougnie, D. Introducing ART: a new method of testing auditory memory with circular reproduction tasks. *Behavior Research Methods, 56*(8), 8330-8348. https://doi.org/10.3758/s13428-024-02477-2 2024

Guérin, S. M. R., Tarlao, C., & **Karabay, A.** Our Journey as Intern Junior Editors at the Journal of Experimental Psychology: Human Perception and Performance. *Science Editing, 11*(2), 160-163. https://doi.org/10.6087/kcse.345 2024

Kandemir, G.*, Wolff, M. J. *, **Karabay, A.**, Stokes, M. G., Axmacher, N., & Akyürek, E. G. Concurrent maintenance of both veridical and transformed working memory representations. *Imaging Neuroscience*, *2*, 1–20. https://doi.org/10.1162/imag_a_00173 2024

Wang, S., **Karabay, A.,** & Akyürek, E. G. Attentional blur and blink: Effects of adaptive attentional scaling on visual awareness. *Consciousness and Cognition, 117*(3), 1–15. https://doi.org/10.1016/j.concog.2023.103627 2024

Altınok, A., **Karabay, A.**, Balta, G, de Jonge, J., & Akyürek, E. G. Effects of gamma-aminobutyric acid on working memory and attention: A randomized, double-blinded, placebo-controlled, crossover trial. *Journal of Psychopharmacology, 37*(6), 554–565. https://doi.org/10.1177/02698811231161579 2023

Chen, I. Y., **Karabay, A.**, Mathôt, S., Bowman, H., & Akyürek, E. G. Concealed identity information detection with pupillometry in rapid serial visual presentation. *Psychophysiology*, *60*(1). 1-14. https://doi.org/10.1111/psyp.14155 2023

Karabay, A., Wilhelm, S. A., de Jonge, J., Wang, J., Martens, S., & Akyürek, E. G. Two faces of perceptual awareness during the attentional blink: Gradual and discrete. *Journal of Experimental Psychology: General, 151*(7), 1520–1541. https://doi.org/10.1037/xge0001156 2022

Altınok, A., **Karabay, A.**, & Akyürek, E. G. Acute effects of cocoa flavanols on visual working memory: Maintenance and updating. *European Journal of Nutrition, 61*(3), 1665–1678. https://doi.org/10.1007/s00394-021-02767-x 2022

Karataş, O., **Karabay, A.**, & Alıcı, T. The effects of long-term cocoa flavanols intake on cognitive functions and mood, and the physiological mechanisms underlying these effects: A literature 2022

review. *Turkish Psychological Articles, 25*(49). 1–26. https://doi.org/10.31828/tpy1301996120211203m000043

Karabay, A., & Kurylo, D. D. Discriminating global orientation of two element sets. *Journal of Cognition, 3*(1), 1-13. https://doi.org/10.5334/joc.127 2020

Karabay, A., & Akyürek, E. G. Temporal integration and attentional selection of color and contrast target pairs in rapid serial visual presentation. *Acta Psychologica*, *196*, 56–69. https://doi.org/10.1016/j.actpsy.2019.04.002 2019

Karabay, A., Saija, J. D., Field, D. T., & Akyürek, E. G. The acute effects of cocoa flavanols on temporal and spatial attention. *Psychopharmacology, 235*(5), 1497–1511. https://doi.org/10.1007/s00213-018-4861-4 2018

Karabay, A., & Akyürek, E. G. The effects of Kanizsa contours on temporal integration and attention in rapid serial visual presentation. *Attention, Perception, & Psychophysics, 79*(6), 1742–1754. https://doi.org/10.3758/s13414-017-1333-6 2017

Book chapters

Karabay, A. Duyum ve algı (Sensation and Perception). In. G. ZEREN (Eds.), *Fizyolojik Psikoloji (Physiological Psychology)* (pp. 203–233). Eğiten Kitap. Ankara, Turkey. 2022

Cotton, R. (2013). *Learning R* (Sünbül, Ö., Sünbül, S. Ö., & **Karabay, A.**, Trans). Pegem Press. İstanbul, Turkey. 2021

PRESENTATIONS

Talks

Invited talks

Karabay, A. Taking a snapshot of the brain: Visual impulse reveals (non)persistent neural activity by perturbation and increasing signal-noise ratio. *CCN meeting, Ghent University* in Gent, Belgium. 08/2022

Conference talks

Karabay, A., Vriens, P., Adams, J., Tran, N., & Sahan, M. I. How you encode is how you remember: Center-to-surround inhibition during encoding underlies repulsion bias in working memory. *19th European Workshop on Imagery and Cognition (EWIC 2026)* in Leiden, the Netherlands. 06/2026

Karabay, A., & Fougnie, D. Revisiting visual awareness: No evidence for levels of processing. *The 25th Vision Sciences Society Meeting (VSS)* in Florida, USA. 05/2025

Chen, I., Mathôt, S., van der Mijn, R., **Karabay, A.**, Bowman, H., & Akyürek, E. G. Concealed familiar face detection with EEG in rapid serial visual presentation. *The 23rd Conference of the European Society for Cognitive Psychology (ESCoP)* in Porto, Portugal. 09/2023

Wang, S., **Karabay, A.**, & Akyürek, E. G. Attentional blur and blink: Effects of adaptive attentional scaling on visual awareness. *The 23rd Conference of the European Society for Cognitive Psychology (ESCoP)* in Porto, Portugal. 09/2023

Akyürek, E. G., Kandemir, G., Wolff, M., **Karabay, A.**, & Wilhelm, S. Probing latent working memory representations in EEG. *The 22nd Conference of the European Society for Cognitive Psychology (ESCoP)* in Lille, France. 08/2022

Chen, I., Buchel, P., **Karabay, A.**, Mathôt, S., & Akyürek, E. G. Concealed familiar face detection with pupillometry in rapid serial visual presentation. *The 22nd Conference of the European Society for Cognitive Psychology (ESCoP)* in Lille, France. 08/2022

Wang, S., **Karabay, A.**, & Akyürek, E. G. Which working memory model accounts best for target representation during the attentional blink? *The 22nd Conference of the European Society for Cognitive Psychology (ESCoP)* in Lille, France. 08/2022

Karabay, A., Wolff, M. J., Ruuskanen, V., & Akyürek, E. G. A visual impulse reveals memoranda embedded in functional connectivity: Evidence for activity-silent WM states. Symposium 05/2022

Interactions between sensory representations and cognitive control in working memory at the *International Conference of Cognitive Neuroscience 2020 (ICON)* in Helsinki, Finland.

Karabay, A., Wang, J., Wilhelm, S., Martens, S., & Akyürek, E. G. Two faces of the attentional blink: Gradual and discrete loss of perceptual awareness. *The 17th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 12/2019

Karabay, A., Wang, J., Martens, S., & Akyürek, E. G. The attentional blink: Binary or gradual? *The 21st Conference of the European Society for Cognitive Psychology (ESCoP)* in Tenerife, Spain. 09/2019

Kandemir, G., **Karabay, A.**, & Akyürek, E. G. Decoding sensory and abstract information from activity silent brain states. *The 21st Conference of the European Society for Cognitive Psychology (ESCoP)* in Tenerife, Spain. 09/2019

Karabay, A., Saija J., Field, D. T., & Akyürek, E. G. The acute effects of cocoa flavanols on temporal and spatial attention. *The 16th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 12/2017

Er N., **Karabay, A.**, Kaynar, G., Uysal, M. M., & Boyraz, F. U. İstemli & istemsiz otobiyografik anılarda yaşa bağlı değişimler (Variations of voluntary and involuntary autobiographical memories depending on age). *The 17th National Psychology Congress* in Istanbul, Turkey. 05/2012

Karabay, A., & Çakmak, M. A. Engellenme karşısında verilen tepkiler ile kişilik tiplerinin ve yükleme çeşitlerinin ilişkisi (The relationship between frustration responses, personality types, and attribution). *The 14th Ulusal Psikoloji Ogrencileri Kongresi (The 14th National Conference of Psychology Students)* in Istanbul, Turkey. 12/2009

Posters

Kalanoski, S., **Karabay, A.**, Chunharas, C., & Fougnie, D. From repulsion to attraction: Object priority drives distinct memory biases in visual working memory. *The 47th European Conference on Visual Perception (ECVP)* in Mainz, Germany. 08/2025

Karabay, A., & Fougnie, D. Evidence against levels of processing theories of visual awareness. *Asia Pacific Conference on Vision (APCV)* in Singapore. 07/2024

Sasin, E., Zhou, Y., **Karabay, A.**, Shrestha, S., & Fougnie, D. Where's Waldo in the mind: Accessing perceptual and semantic attributes in perception and working memory. *The 24th Vision Sciences Society Meeting (VSS)* in Florida, USA. 05/2024

Chen, I., Mathôt, S., van der Mijn, R., **Karabay, A.**, Bowman, H., & Akyürek, E. G. Concealed familiar face detection with EEG in rapid serial visual presentation. *The 45th European Conference on Visual Perception (ECVP)* in Paphos, Cyprus. 08/2023

Fougnie, D., **Karabay, A.**, Nijenkamp, R., & Sarampalis, A. Introducing ART: a new method of testing auditory memory with circular reproduction tasks. *The 23rd Vision Sciences Society Meeting (VSS)* in Florida, USA. 05/2023

Chen, I., **Karabay, A.**, Mathôt, S., Buchel, P., van der Mijn, R., Bowman, H., & Akyürek, E. G. Concealed familiar face detection with oculomotor measures and EEG in rapid serial visual presentation. *The 23rd Vision Sciences Society Meeting (VSS)* in Florida, USA. 05/2023

Karabay, A., Kandemir, G., Kovács, E.R., & Akyürek, E. G. Pinging attentional task rules: Neural mechanisms underlying attending and suppressing in visual search tasks. *The 22nd Conference of the European Society for Cognitive Psychology (ESCoP)* in Lille, France. 08/2022

Altınok, A., **Karabay, A.**, Lorist, M. M., Weiden, D., & Akyürek, E. G. The acute effects of caffeine and cocoa flavanols on working memory and attention. *The 22nd Conference of the European Society for Cognitive Psychology (ESCoP)* in Lille, France. 08/2022

Chen, I., Buchel, P., **Karabay, A.**, Mathôt, S., Bowman, H., & Akyürek, E. G. Concealed information detection with pupillometry in rapid serial visual presentation. *The International Conference of Cognitive Neuroscience 2020 (ICON)* in Helsinki, Finland. 05/2022

Karabay, A., Wolff, M. J., Ruuskanen, V., & Akyürek, E. G. Visual impulse perturbation: Neural sonar or just noise-reducer? *The 18th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 04/2022

Chen, I., **Karabay, A.**, Mathôt, S., Bowman, H., & Akyürek, E. G. Concealed identity information detection with pupillometry in rapid serial visual presentation. *The 18th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 04/2022

Wang, S., **Karabay, A.**, & Akyürek, E. G. (2022, April). Which working memory model accounts best for target representation in the attentional blink? *The 18th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 04/2022

Altınok, A., Balta, G., **Karabay, A.**, & Akyürek, E. G. The effects of gamma-aminobutyric acid (GABA) on working memory and attention. *The 18th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 04/2022

Kandemir, G., **Karabay, A.**, & Akyürek, E. G. Seeing double? Activity-silent maintenance of mental rotation products and their originals. *The 17th winter conference of De Nederlandse Vereniging voor Psychonomie (NVP)* in Egmond, the Netherlands. 12/2019

Altınok, A., **Karabay, A.**, & Akyürek, E. G. The acute effects of cocoa flavanols on visual working memory. *The 21st Conference of the European Society for Cognitive Psychology (ESCoP)* in Tenerife, Spain. 09/2019

Kandemir, G., **Karabay, A.**, & Akyürek, E. G. Decoding visual working memory before and after mental operations. *The BCN Winter Meeting* in Twente, the Netherlands. 02/2019

Karabay, A., & Akyürek, E. G. Target color and contrast influences temporal attention in rapid serial visual presentation. *The International meeting of the Psychonomics Society in Amsterdam*, Netherlands; *The 40th European Conference on Visual Perception (ECVP)* in Berlin, Germany. 08/2017

Karabay, A., & Akyürek, E. G. Kanizsa effects on temporal integration and attention. *The Heymans Symposium* in Groningen, the Netherlands. 04/2017

Karabay, A., & Kurylo, D. D. Perceptual grouping by orientation coherence. *The Science Day, Brooklyn College* in New York, USA. 05/2015

Er, N., **Karabay, A.**, Kaynar, G., Uysal, M. M., & Boyraz, F. U. Autobiography of aging mind: age-related effects on voluntary and involuntary autobiographical memories. *The IX International Cognitive Neuroscience Meeting* in Istanbul, Turkey. 04/2012

SUPERVISED PhD THESES

Wang, S. *Temporal Constraints in Visual Cognition Explored with Rapid Serial Visual Presentation.* University of Groningen, the Netherlands. https://doi.org/10.33612/diss.1529308598 2026

SUPERVISED MASTER THESES

‡ indicates published projects

‡Ruuskanen, V. *Investigating the functionality of visual impulses in decoding working memory using matching and non-matching impulse stimuli.* University of Groningen, the Netherlands. 2022

Bapoğlu, O. A. *Pinging individual features of multi-feature objects: an EEG study.* University of Groningen, the Netherlands. 2021

‡McAllister, C. *Effect of cocoa flavanols on attention.* University of Groningen, the Netherlands. 2017

RELEVANT SKILLS

Technical skills: JASP (*advanced*); SPSS (*advanced*); R (*advanced*); Matlab (*moderate/advanced*); Python (*moderate/advanced*); OpenSesame (*advanced*); E-prime (*advanced*); MS Office (*advanced*); Memory modeling; EEG; Multivariate Pattern Analysis; Canvas; Blackboard; GitHub; Classification; WordPress

Language skills: Turkish (*native*); English (*professional*); Dutch (*beginner*)

GRANTS & FELLOWSHIPS

Travel Grant from University of Birmingham for attendance to VSS 2025 **(1,400 £).** 2025

YLSY Fellowship from Ministry of National Education, Turkey. Complete coverage of tuition, stipend, and partial coverage of research costs for masters and Ph.D. degrees (**200,000$**). 2011–2019

AWARDS

Open research award for the study 'Concealed familiar face detection with pupillometry in rapid serial visual presentation,' by Open Science Programme, University of Groningen, Groningen, the Netherlands. 2022

Best poster presentation award in Searching for the Working Memory Engram workshop with the presentation 'A visual impulse reveals memoranda embedded in functional connectivity: Evidence for activity-silent WM states', Groningen, the Netherlands. 2022

ORGANIZED WORKSHOP & SEMINARS

UoBD Psychology Conference 2026: Redefining Student Mental Health and Wellbeing, University of Birmingham Dubai, UAE. 09/2026

Searching for the Working Memory Engram, University of Groningen, the Netherlands. Organized an international hybrid workshop with prof. Elkan G. Akyürek. 05/2022

Introduction to Python/OpenSesame, Leibniz Institute for Psychology, Germany. Worked as a tutor for the workshop by dr. Sebastiaan Mathôt. 08/2020

Mindwise Science Communication Workshop, University of Groningen, the Netherlands. Organized a workshop with the Mindwise Science Communication PhD Committee. 2019; 2020

ORGANIZATIONAL EXPERIENCE

Post-Graduate Researcher Lead, School of Psychology, University of Birmingham Dubai, Dubai 2025–Present

Ph.D. and postdoc representative, Department of Experimental Psychology, University of Groningen, the Netherlands. 2021–2022

Mind-wise science communication Ph.D. committee member, University of Groningen, the Netherlands. 2018–2021

Psychology Ph.D. council member, University of Groningen, the Netherlands. 2016–2019

PEER REVIEW & EDITORIAL CONTRIBUTIONS

Editorial experience

Junior editor, Journal of Experimental Psychology: Human Perception and Performance 2023–2024
Supervisor: Prof. Isabel Gauthier

Grant Reviews

National Science Foundation (NSF), USA 2024; 2025

Journal Reviews

Attention, Perception, & Psychophysics

Acta Psychologica

eNeuro

Journal of Articles in Support of the Null Hypothesis

Journal of Cognition

SELECTED CERTIFICATES

Teaching and mentoring in academic learning communities, Teaching Unit Psychology of University of Groningen, Groningen, the Netherlands. 2018

Multivariate pattern analyses (MVPA) in EEG/MEG, Experimenteel-Psychologische Onderzoekschool, Amsterdam, the Netherlands. 2018

Autumn school: Attention, Experimenteel-Psychologische Onderzoekschool, Amsterdam, the Netherlands. 2018